# The Three Butter Truth

## **Chyura Butter** (Diploknema butyracea), **Shea Butter** (Vitellaria paradoxa) **& Kokum Butter** (Garcinia indica)

*Comparative assessment of three natural seed butters for beauty-product development on basis of physico-chemical characteristics*

**Prepared by:** Aadvwaya Passi, Class 12th
**Date:** April - September 2026
Under the guidance of : Dr. Antaryami Jena
Chief Scientist, Lotus Herbals Pvt Ltd., New Delhi, INDIA
Academic year: 2026–27

## CERTIFICATE

This is to certify that the research paper titled 'THE THREE BUTTER TRUTH' has been completed by Aadvwaya Passi of class 12th, The Shri Ram School, Moulsari, Gurugram, Haryana, INDIA, under my guidance. The work is based on laboratory testing of the butters at the Lotus Herbals R&D laboratory, a skin-hydration study conducted by the student, and the student's own understanding, lab tests and analysis of the lab data.

Signature of the Chief Mentor: _____ Date: __September 2026__

Signature of the Student: _____

## Table of Contents


1. INTRODUCTION ..... 4
2. AIM AND OBJECTIVES ..... 4
3. ABOUT THE THREE BUTTERS ..... 5
4. THE TESTS USED IN THIS STUDY AND WHAT THEY MEAN ..... 5
5. OBSERVATIONS: PHYSCIO-CHEMICAL DATA ..... 6
6. OBSERVATIONS: COMPARISON AND GRAPHS ..... 7
7. DISCUSSION: WHAT THE NUMBERS MEAN FOR BEAUTY PRODUCTS ..... 9
8. WHICH BUTTER SUITS WHICH PRODUCT ..... 11
9. ADVANTAGES OF CHYURA BUTTER TO LOTUS HERBALS ..... 12
10. MOISTURISATION EFFICACY STUDY (EXPERIMENT) ..... 14
11. CONCLUSION ..... 19
12. BIBLIOGRAPHY ..... 20




## 1. Introduction

When we rub a body butter or a lip balm on our skin, the soft, creamy part of it is usually a "plant butter". A plant butter is a fat taken out of a seed or nut that stays solid at room temperature but melts when it touches warm skin. The most famous one is shea butter, which comes from Africa and is imported into india. Two Indian plant butters are also used by beauty companies: **Kokum butter** from the Western Ghats and **Chyura butter** (also called chiuri or phulwara butter) from the Himalayan hills.

These three butters look alike in a jar, but their chemistry is different, and that decides how they feel on the skin, how long they stay fresh, and what kind of product they are good for. Chemists check a fat with a few simple laboratory tests: saponification value, acid value, iodine value, peroxide value, melting point and moisture content.

My reason for choosing this topic is simple. India imports most of its shea butter from West Africa, while chyura trees grow in our own hills and support poor tribal families. If chyura butter is as good as shea, Indian beauty brands have a strong reason to use it.

## 2. Aim and Objectives

**Aim:**

To compare the physico-chemical properties of chyura, shea and kokum butter and find out which one is most effective in a oil in water moisturizer (O/W Emulsion) and find out suitability to Indian beauty brands.

**Objectives:**

1. To understand what each laboratory test (saponification value, acid, iodine, peroxide value, melting point) tells us about a fat.
2. To compare the three butters using tables and graphs.
3. To explain what the differences mean for making skin creams / moisturisers.
4. To give reasons why an Indian beauty brands should prefer the domestically grown chyura butter.



## 3. About the three butters

| | Chyura butter | Shea butter | Kokum butter |
|---|---|---|---|
| Plant | Diploknema butyracea (Indian butter tree, chiuri, phulwara) | Vitellaria paradoxa (shea tree, karite) | Garcinia indica (kokum, wild mangosteen) |
| Family | Sapotaceae | Sapotaceae | Clusiaceae |
| Where it grows | Sub-Himalayan hills of Uttarakhand (Pithoragarh, Kumaon), Nepal, Sikkim, North-East India | Savanna belt of West and East Africa (Ghana, Burkina Faso, Nigeria, Mali, Uganda) | Western Ghats: Konkan coast of Maharashtra, Goa, coastal Karnataka, India |
| Part used | Seed kernel (about 50–65% fat) | Nut kernel (about 45–55% fat) | Seed kernel (about 33–45% fat) |
| Season | Fruit ripens June–July | Fruit comes May–August | Seeds September–December |
| How it is made | Seeds dried, roasted, pounded or pressed; fat skimmed; called "chiur ghee" | Nuts boiled, dried, crushed, kneaded with water; fat skimmed | Seeds dried, cracked, kernels pressed |
| Looks like | Light brown to pale yellow soft fat; white when refined | Ivory to yellow soft fat | Light grey to pale yellow hard fat |
| Who collects it | Van-Raji and other hill tribes of Uttarakhand, India; Chepang community in Nepal | Mostly rural women in Africa ("women's gold") | Konkan farmers and collectors in South and Western India |

*Table 1: Basic facts about the three butters*

All three are "tree-borne oilseeds", which means the fat comes from the seed of a wild or semi-wild tree, not from a farm crop. The chyura tree is special because it also gives honey (bees love its flowers), edible fruit, fodder and fuel, so the whole tree is useful to hill families. The Dehradun scientists collected their chyura seeds from the forests of Pithoragarh district in Uttarakhand, India.

## 4. The tests used in this study and what they mean

Before looking at the numbers, it helps to know what each test measures. These are the standard tests that any oil or fat laboratory does.



| Test | What is done | What the number tells us | Good for a beauty product? |
|---|---|---|---|
| Saponification value (SV) | The fat is boiled with potassium hydroxide (KOH). The amount of KOH used up (mg per g of fat) is measured. | How much alkali is needed to turn the fat into soap. A higher SV means shorter fatty-acid chains. | Important for soap making. For creams it mainly shows the type of fatty acids present. |
| Acid value (AV) | The fat is dissolved and titrated with KOH to neutralise free acids. | How many fatty acids have broken loose from the fat. A high AV means the fat is old, roughly handled, or unrefined. | Lower is better. A high AV fat smells, can irritate skin and spoils faster. Refining brings it down. |
| Iodine value (IV) | Iodine is added; the fat absorbs iodine at its double bonds. The amount absorbed (g per 100 g) is measured. | How "unsaturated" the fat is. A low IV means mostly saturated fat, which is more stable and more solid. | A low IV is good for shelf life. A very low IV makes the fat hard and waxy. |
| Peroxide value (PV) | Measures peroxides that form when a fat reacts with oxygen (meq O2 per kg). | How much the fat has already started to go rancid. | Lower is better. Fresh fats are below 10. |
| Melting point | The fat is heated slowly and the temperature at which it turns liquid is noted. | Whether the butter will be soft or hard, and whether it melts on the skin (skin is about 32–35 °C). | Around 35–42 °C is ideal: solid in the jar, melts on skin. |
| Moisture content | The fat is heated and the loss of weight is measured. | How much water is trapped in the fat. | Lower is better; water allows germs to grow. |

*Table 2: Meaning of the laboratory tests*

## 5. Observations: Physcio-chemical data

All three butters were tested in the Quality Control laboratory of Lotus Herbals Pvt Ltd., New Delhi, by the same analysts and methods.

| Parameter | Kokum (refined) – observed | Shea – observed | Chyura – observed |
|---|---|---|---|
| Appearance | Off-white solid fat | Solid | Opaque solid fat |
| Free fatty acid (%) | 0.06 | 0.15 | 2.8 |
| Acid value (mg KOH/g) | 0.4 | 0.3 | 5.6 |
| Melting point (°C) | 32.5 | 33 | 38 |
| Unsaponifiable matter (%) | 1.2 | 1.4 | 2.15 |
| Iodine value (g $I_2$/100 g) | 40 | 45 | 39.75 |
| Saponification value (mg KOH/g) | 186.5 | 185 | 195.55 |
| Peroxide value (meq/kg) | 0.6 | 1.0 | 0.4 |
| Moisture / volatile matter, oven (%) | 0.095 | 0.032 | 0.059 |

*Table 3: Lotus Herbals QC laboratory results for the three butters, with the company's specifications for kokum and shea butter*

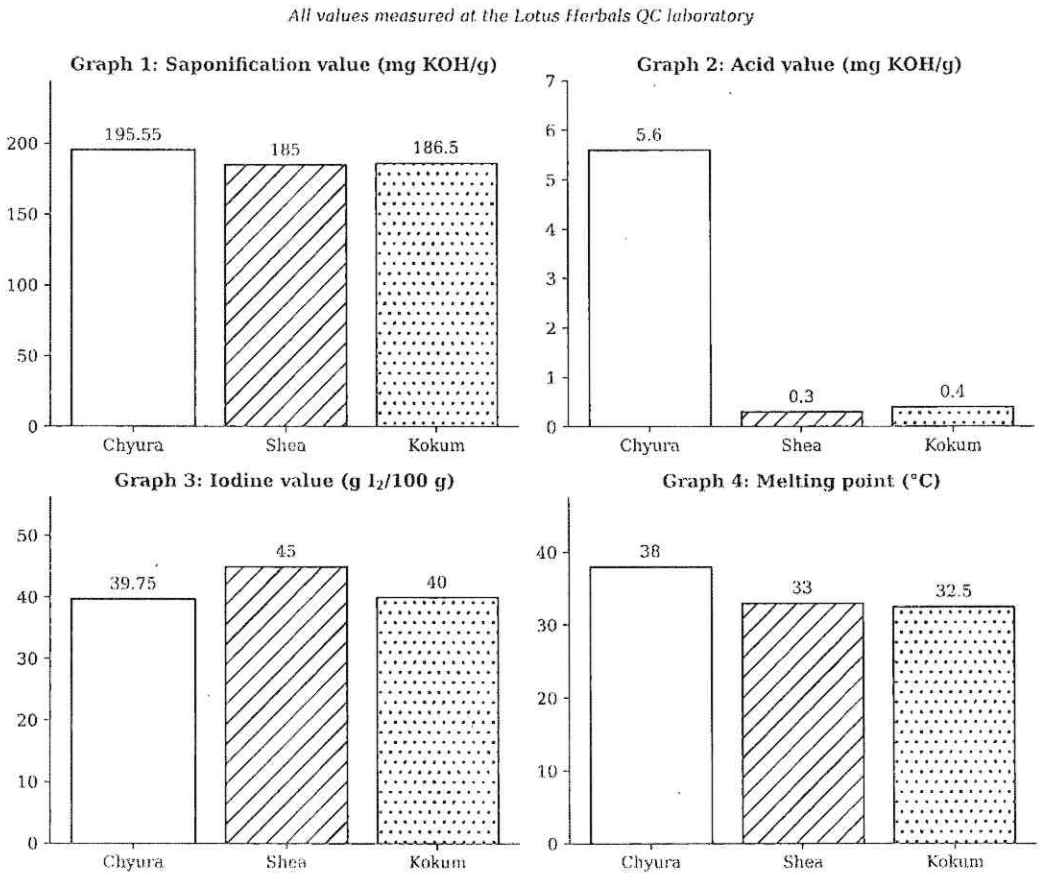
## 6. Observations: Comparison and graphs

| Property | Chyura (lab) | Shea (lab) | Kokum (lab) | Which is best for skin care? |
|---|---|---|---|---|
| Melting point (°C) | 38 | 33 | 32.5 | All three melt on skin. Chyura (38 °C) is the firmest of the three samples; the refined shea and kokum samples are softer (33 and 32.5 °C). |
| Saponification value | 195.55 | 185 | 186.5 | All normal for a C16–C18 fat. Chyura's SV is the highest, which points to shorter average chains (more palmitic acid). |
| Acid value (crude) | 5.6 (R&D sample, partly refined) | 0.3 (refined) | 0.4 (refined) | The chyura R&D sample was only partly refined. Fully refined chyura is expected to be below 1, like the others. |

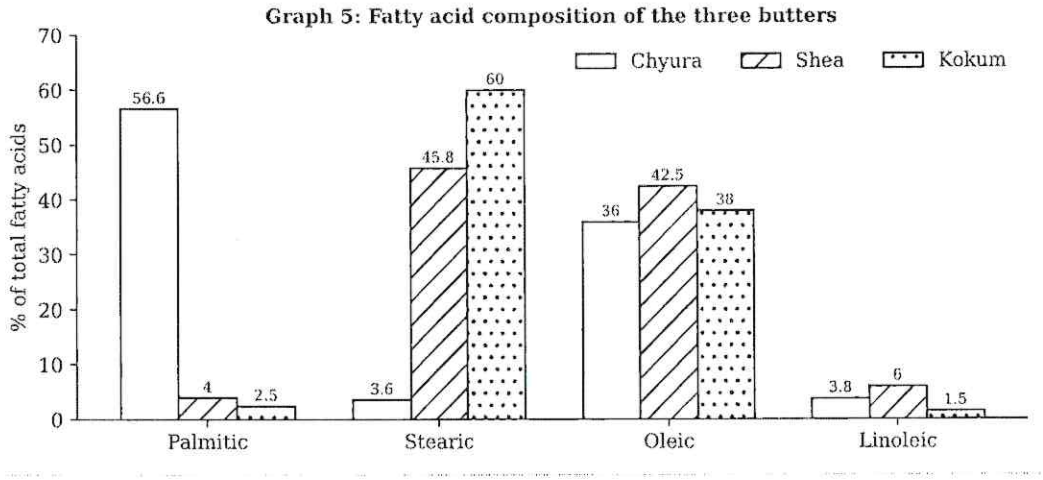


| Property | Chyura (lab) | Shea (lab) | Kokum (lab) | Which is best for skin care? |
|---|---|---|---|---|
| Iodine value | 39.75 | 45 | 40 | Chyura (39.75) and kokum (40) are the most saturated; shea (45) has the most double bonds, so chyura should resist rancidity best. |
| Peroxide value | 0.4 | 1.0 | 0.6 | All fresh and good quality; chyura's value is the lowest. |
| Moisture | 0.059% | 0.032% | 0.095% | All three well below 0.1%; no moisture problem in any sample. |
| Dominant fatty acid | Palmitic + oleic | Stearic + oleic | Stearic (~60%) | Chyura is a "palmitic butter", unlike the other two "stearic butters". |

*Table 4: Side-by-side comparison of the three butters (all values measured at the Lotus Herbals QC laboratory)*




*Graphs 1–4: Comparison of the four main chemical values (all three butters measured in the same laboratory)*




*Graph 5: Typical fatty-acid composition of the three butters*

Observations from the graphs:

5. Graph 1: All three saponification values are in the normal range for a cosmetic fat. Chyura (195.55) is the highest, shea (185) the lowest.
6. Graph 2: The chyura R&D sample has a much higher acid value (5.6) than the refined shea (0.3) and kokum (0.4), though far lower than crude village butters. This is the one weakness in the chyura data and is explained in the discussion.
7. Graph 3: Chyura has the lowest iodine value (39.75), marginally below kokum (40) and clearly below shea (45), so it has the fewest double bonds and should be the most stable against air and heat.
8. Graph 4: All three melt between 32.5 and 38 °C, at or just above skin temperature, so all three "melt on touch". Chyura is the firmest of the three samples.
9. Graph 5: Chyura stands apart because more than half of it is palmitic acid, while shea and kokum are built on stearic acid.

## 7. Discussion: what the numbers mean for beauty products

### 7.1 Stability and shelf life

A beauty product sits on a shelf for one to three years, so the fat inside must not go rancid (oxidise). Two numbers tell us about this: the iodine value and the peroxide value. Chyura has the lowest iodine value of the three (39.75), marginally below kokum (40) and clearly

Lotus Herbals Pvt. Ltd. Sultanpur New Delhi Forest Lane

below shea (45), so it has the fewest carbon–carbon double bonds, the weak spots where oxygen attacks. Its peroxide value of 0.4 is also the lowest (kokum 0.6, shea 1.0), so all three samples were fresh and chyura the freshest. Shea, with the most double bonds, is the least stable of the three and is well known to go grainy and rancid if stored badly. On stability, chyura scores best, kokum close behind, and shea last.

## 7.2 The explanation for the acid value of chyura

The chyura sample had an acid value of 5.6 mg KOH/g, compared with 0.3 for the refined shea and 0.4 for the refined kokum. The sample tested was a partly refined R&D sample, while the shea and kokum samples are refined, factory-grade butters. Crude village butters of any kind carry much higher acid values (crude shea is allowed up to 8% free fatty acid, an acid value of about 16), and when a fat is refined the free fatty acids are neutralised and washed out and the acid value drops below 2. So the comparison is between a partly finished sample and finished ones. Chyura must be bought in a refined, cosmetic grade with a certificate of analysis, exactly as beauty companies already do for shea and kokum.

## 7.3 Texture and feel

The melting points of the three samples range from 32.5 °C (kokum) and 33 °C (shea) to 38 °C (chyura), all at or just above skin temperature, but the fatty acids are different, and that changes the feel. Kokum, with about 60% stearic acid, is usually hard and crumbly and gives a dry, light finish; the refined kokum sample tested here, however, melted at only 32.5 °C. Shea is soft and rich. Chyura, with about 57% palmitic acid and 36% oleic acid, is soft, smooth and cushiony, more like ghee, and with the highest melting point of the three it holds its shape best in the jar. Palmitic acid is also a natural part of our own skin oil (sebum), so chyura's main fatty acid is one the skin already knows.

Shea butter is usually valued for its unsaponifiable matter, which contains triterpene esters that have been shown to calm inflammation and even boost sunscreen performance. In the samples tested here, however, the unsaponifiable content was 1.4% for shea, 1.2% for kokum and 2.15% for chyura, so this refined shea sample did not carry an unusually large active fraction, and chyura's was the highest of the three (though part of that may be incomplete refining). So the usual "shea wins on active chemistry"

argument is weaker than expected for refined stock, while chyura and kokum win on stability and chyura on firmness.

### 7.4 Saponification value and soap

All three saponification values are in the range of fats that make a good, hard soap, and chyura's (195.55) is the highest of the three. In Nepal, chyura butter has been used for soap for generations, and the seed kernel also contains natural saponins, which help the lather. So chyura is a natural choice for soap and body bars even in its crude form.

## 8. Which butter suits which product

| Product | Chyura | Shea | Kokum | Remarks |
|---|---|---|---|---|
| Body butter / winter cream | Very good | Very good | Good | Chyura's soft, ghee-like feel and stability suit this best |
| Lip balm | Good | Good | Very good | Kokum gives a firm stick; chyura adds softness |
| Foot and hand repair cream | Very good | Very good | Very good | All three are traditional cracked-skin remedies |
| Face cream | Needs testing | Good | Very good | Chyura's pore-clogging rating is not yet tested |
| Soap / body bar | Very good | Good | Good | Chyura has a long soap tradition |
| Sunscreen | Needs testing | Very good | Good | Shea is a proven SPF booster; chyura has no sun-care data yet |
| Hair butter / conditioner | Good | Very good | Fair | Chyura is traditionally used for dry hair in Nepal |
| Hot-climate products (sticks, balms) | Fair | Poor alone | Very good | Kokum keeps its shape in Indian summers |

*Table 6: Suitability of each butter for common beauty products*

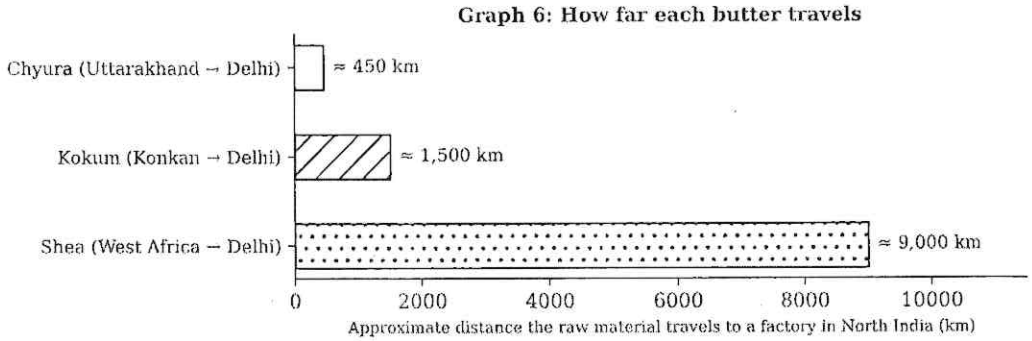
The ratings in Table 6 use skin hydration in a simple cream (Section 10) as an approximate proxy for how each butter performs as the emollient in other leave-on products, together with the chemistry above. A sensible approach for a brand is to use the butters together: chyura for its soft feel and story, kokum to add hardness in summer products, and a small amount of shea only where its sun-protection boost is needed.

## 9. Advantages of chyura butter to Lotus Herbals

From the chemistry alone, the three butters are close enough that all of them can be used in beauty products. The real difference comes from where they come from and who benefits. Below are the reasons why Lotus Herbals, as a North-Indian herbal beauty company, should prefer chyura.

### 9.1 It is grown in India, not imported

Shea trees grow only in Africa. Every kilogram of shea used in India is imported, shipped about 9,000 km, paid for in foreign currency, and exposed to African weather, export rules and price swings (shea prices moved between about US$3 and US$10 per kg in recent years). Chyura grows in Uttarakhand, Sikkim and the North-East, a few hundred kilometres from Lotus Herbals' factories in North India. Kokum is also Indian, but it grows only on the Konkan coast, so for a North Indian company chyura is the closest source of all.




*Graph 6: Approximate distance each raw material travels to a factory in North India*

### 9.2 It supports Indian hill communities

The chyura tree is the main cash crop of the Van-Raji tribe of Pithoragarh, one of India's most vulnerable forest communities, and of the Chepang people across the border in Nepal. Studies show that most of the possible chyura butter is never collected because there is no steady buyer. A beauty brand that signs supply contracts with these villages creates income at home, in line with "Vocal for Local" and "Make in India", instead of sending money to African exporters.

### 9.3 It is chemically just as good, and in some ways better

10. Lowest iodine value (39.75) of the three, so the best natural resistance to going rancid.
11. Peroxide value (0.4) lower than fresh cow ghee.
12. Melting point (38 °C) in the ideal "melts on skin" range.
13. Soft, smooth, ghee-like texture with palmitic acid, a fatty acid the skin already contains.
14. No graininess problem like shea.
15. Highest skin hydration of the three butters at 6% in the Corneometer study (Section 10), and the only butter that hydrated better when its level was raised from 4% to 6%.

### 9.4 It gives brands a story no one else has

Shea is in thousands of products and kokum is already used by well-known Indian Ayurvedic brands. Chyura is almost unknown in mainstream Indian beauty. A brand that uses it can tell a fresh story: a "Himalayan butter tree", a rare ingredient, and a tribal community behind it. In marketing, being first with an ingredient is worth a lot.

### 9.5 It is good for the environment

The chyura tree holds soil on steep Himalayan slopes, feeds honeybees, and grows on land where farming is hard. Buying its seeds gives villagers a reason to protect the trees instead of cutting them for firewood. Shea trees, on the other hand, are listed as "Vulnerable" by the IUCN because of over-cutting in Africa.

### 9.6 What Lotus Herbals must still do

To be fair, chyura has two gaps that a brand must close: it must be refined to bring the acid value down and remove saponins, and the six-hour hydration study in Section 10 was done on only two volunteers. Both are solvable. Refining is a standard factory step, and a four-week skin-hydration study on a full panel would give the brand its own clinical data, something no competitor has for chyura.

## 10. Moisturisation efficacy study (experiment)

### 10.1 Objective

To measure skin hydration on the volar forearm after applying six test moisturising creams, each containing chyura, shea or kokum butter at 4% or 6%, over six hours against an untreated control site, and so to find

(a) which butter hydrates best

(b) which of the two concentrations is the better choice.

Hydration was measured with a Corneometer, which reads the electrical capacitance of the skin surface; the higher the reading, the more water the stratum corneum holds (dry skin below 20, normal 20–40, well hydrated above 40).

### 10.2 Test products

| Code | Test product | Butter level (% w/w) |
|---|---|---|
| Control | Untreated site | 0 |
| CH4 / CH6 | Moisturising cream with chyura butter | 4 / 6 |
| SH4 / SH6 | Moisturising cream with shea butter | 4 / 6 |
| KO4 / KO6 | Moisturising cream with kokum butter | 4 / 6 |

*Table 7: Test products. All six creams share the same oil-in-water base; only the butter and its level differ.*

### 10.3 Apparatus

| S. No. | Apparatus | Least count |
|---|---|---|
| 1. | Beaker 250 ml | 50 ml |
| 2. | Beaker 500 ml | 100 ml |
| 3. | Corneometer (Courage + Khazaka company, MPA 6 model) | N.A. |
| 4. | Stirrer | N.A. |
| 5. | Weighing balance | 0.001 g |
| 6. | Melting plate | N.A. |

*Table 8: Apparatus used in the experiment*

### 10.4 Procedure

1. The six creams were prepared from one common oil-in-water base.

2. Two healthy adult volunteers (coded PK and NK) took part. The volar forearms of each participant were cleaned with an alcohol swab.
3. Test sites of 3 cm × 3 cm were marked on the volar forearms: one untreated control site and one site for each test product. Participants were acclimatised for 1 hour at the start of the study in a room kept at 25 °C and 40–60% relative humidity, and stayed in the room for the whole study.
4. Baseline Corneometer readings were taken on the test sites after acclimatisation (0 h).
5. Approximately 0.030 g of each test product was applied to its test site by massaging for 30 seconds with a fingerstall. No product was applied to the control site.
6. Corneometer readings were repeated at 1 h, 2 h, 4 h and 6 h after application.

## 10.5 Results

### Raw Corneometer data

Individual readings (arbitrary units) for the two participants. The baseline was measured once per participant (PK 28.12; NK 30.53) and applied to all sites.

| Test product | Participant | 0 h | 1 h | 2 h | 4 h | 6 h |
|---|---|---|---|---|---|---|
| Control (no product) | PK | 28.12 | 23.1 | 25 | 31 | 25 |
| Control (no product) | NK | 30.53 | 19.48 | 23 | 26 | 25.5 |
| Chyura 4% | PK | 28.12 | 42.62 | 45 | 51 | 54 |
| Chyura 4% | NK | 30.53 | 43.43 | 44 | 58.3 | 54.1 |
| Chyura 6% | PK | 28.12 | 32.65 | 63 | 59 | 59 |
| Chyura 6% | NK | 30.53 | 41.82 | 57 | 61 | 58.1 |
| Shea 4% | PK | 28.12 | 41.78 | 50 | 52 | 55 |
| Shea 4% | NK | 30.53 | 51.37 | 46 | 59 | 54.1 |
| Shea 6% | PK | 28.12 | 25.1 | 56 | 56 | 57 |
| Shea 6% | NK | 30.53 | 44.6 | 48 | 53 | 56.2 |
| Kokum 4% | PK | 28.12 | 37 | 56 | 57 | 55 |
| Kokum 4% | NK | 30.53 | 49 | 51 | 54 | 54.5 |



| Test product | Participant | 0 h | 1 h | 2 h | 4 h | 6 h |
|---|---|---|---|---|---|---|
| Kokum 6% | PK | 28.12 | 28.7 | 57 | 50 | 56 |
| Kokum 6% | NK | 30.53 | 38.5 | 41 | 52 | 55.9 |

*Table A1: Raw Corneometer readings by participant*

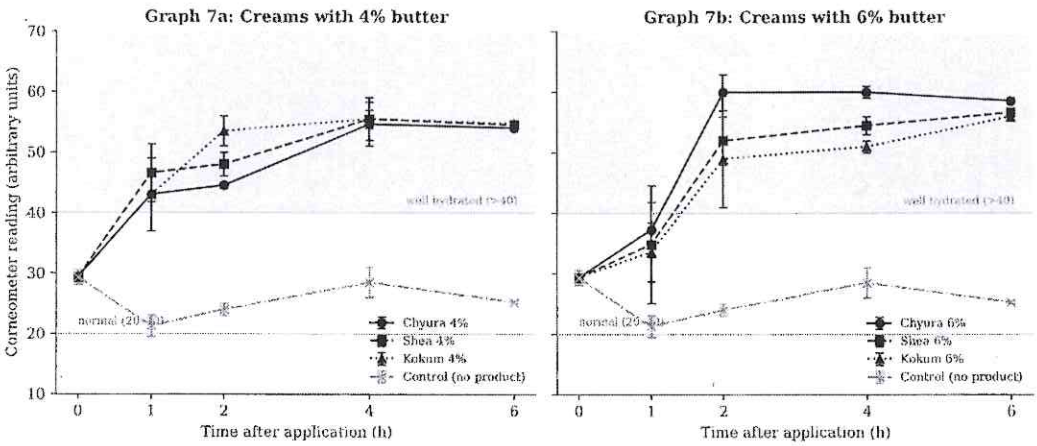
Table 9 gives the mean Corneometer reading of the two participants at each time point, with the half-range between them as the spread. Table 10 gives the change at 6 hours and the net cumulative hydration over the six hours (area under the change-versus-time curve after subtracting the control site). The individual readings are in Appendix A.

| Test product | 0 h (baseline) | 1 h | 2 h | 4 h | 6 h |
|---|---|---|---|---|---|
| Control (no product) | 29.3 ± 1.2 | 21.3 ± 1.8 | 24.0 ± 1.0 | 28.5 ± 2.5 | 25.2 ± 0.2 |
| Chyura 4% | 29.3 ± 1.2 | 43.0 ± 0.4 | 44.5 ± 0.5 | 54.6 ± 3.6 | 54.0 ± 0.1 |
| Chyura 6% | 29.3 ± 1.2 | 37.2 ± 4.6 | 60.0 ± 3.0 | 60.0 ± 1.0 | 58.5 ± 0.5 |
| Shea 4% | 29.3 ± 1.2 | 46.6 ± 4.8 | 48.0 ± 2.0 | 55.5 ± 3.5 | 54.5 ± 0.5 |
| Shea 6% | 29.3 ± 1.2 | 34.9 ± 9.8 | 52.0 ± 4.0 | 54.5 ± 1.5 | 56.6 ± 0.4 |
| Kokum 4% | 29.3 ± 1.2 | 43.0 ± 6.0 | 53.5 ± 2.5 | 55.5 ± 1.5 | 54.8 ± 0.2 |
| Kokum 6% | 29.3 ± 1.2 | 33.6 ± 4.9 | 49.0 ± 8.0 | 51.0 ± 1.0 | 56.0 ± 0.1 |

*Table 9: Mean Corneometer reading (arbitrary units) ± half-range between the two participants*

| Test product | Change at 6 h vs baseline | Net change at 6 h vs control | Net cumulative hydration 0–6 h (AUC, units·h) | Rank |
|---|---|---|---|---|
| Chyura 4% | 24.7 | 28.8 | 133.5 | 5 |
| Chyura 6% | 29.2 | 33.3 | 166.2 | 1 |
| Shea 4% | 25.2 | 29.3 | 144.5 | 3 |
| Shea 6% | 27.3 | 31.4 | 138.9 | 4 |
| Kokum 4% | 25.4 | 29.5 | 149.4 | 2 |
| Kokum 6% | 26.6 | 30.7 | 125.5 | 6 |

*Table 10: Change from baseline at 6 h, net change after subtracting the control site, and net cumulative hydration over 0–6 h*




*Graph 7: Corneometer reading against time for the 4% creams (a) and the 6% creams (b), with the untreated control. Error bars show the range between the two participants; the shaded zone is the "well hydrated" band (above 40).*

## 10.6 What the results show

**How chyura compares at 4% and 6%.** At 4% the three butters performed the same: after 6 h the readings were 54.0 (chyura), 54.5 (shea) and 54.8 (kokum), a spread of 0.8 units, which is less than the ±1.6 unit accuracy of the Corneometer at this reading. At 6% chyura moved clearly ahead. It reached 60.0 units by 2 h, while shea was at 52.0 and kokum at 49.0, held 58.5–60.0 units for the rest of the study, and finished at 58.5 against 56.6 (shea) and 56.0 (kokum). Over the full six hours the 6% chyura cream gave a net cumulative hydration of 166.2 units·h, which is 20% more than shea (138.9) and 32% more than kokum (125.5).

**Optimum concentration across all six creams.** Graph 7 and Table 10 show that 6% chyura is the best of the six test creams. It ranks first on net cumulative hydration (166.2 units·h, ahead of 4% kokum at 149.4 and 4% shea at 144.5) and has the highest reading at 6 h (58.5). Chyura was also the only butter to improve when its level went from 4% to 6% (+24% in cumulative hydration), whereas shea stayed level (−4%) and kokum fell (−16%). The reason lies in its chemistry. Chyura is a soft palmitic–oleic fat that melts at 38 °C, so it spreads into a thin, continuous film on the skin that slows water loss, and adding more butter makes this film thicker. The stearic-rich shea and kokum butters seem to reach their limit at 4%, and above that they mainly make the cream heavier and slower to absorb.

**How this connects to the study.** Chyura is a butter that mainstream cosmetics hardly use, yet at 6% it doubled skin hydration (29.3 → 58.5 units, +99.7%) and beat both imported shea (+20%) and kokum (+32%) on cumulative hydration. It was also the only butter still improving at the highest level tested. For Lotus Herbals this means a butter

sourced about 450 km from its factories could replace shea, which travels 9,000 km, in its moisturisers with no loss in performance.

## 10.7 Uncertainties and errors

| Source | Instrument / quantity | Least count or accuracy | Absolute uncertainty | Percentage uncertainty |
|---|---|---|---|---|
| Hydration reading | Corneometer MPA 6 | ± 3% of reading | ± 1.8 units at 58.5; ± 0.8 units at 25.3 (control) | ± 3.0% |
| Between-participant spread (6 h) | Two volunteers | Half-range of PK and NK | ± 0.05 – 0.45 units | ± 0.1 – 0.8% |
| Mass of cream applied | Weighing balance | 0.001 g | ± 0.0005 g on 0.030 g | ± 1.7% |
| Room temperature | Thermometer | 1 °C | ± 0.5 °C on 25 °C | ± 2.0% |

*Table 11: Sources of uncertainty and error in the experiment*

## 10.8 Limitations of the experiment

1. Two participants is too few for any statistical test; a confirmatory study needs at least 10–15 volunteers.
2. A single application over six hours measures immediate hydration, not barrier repair or the effect of daily use over weeks.
3. Water loss through the skin (TEWL) was not measured, so the occlusive-film explanation in 10.6 could not be confirmed directly.
4. Alcohol swabbing produced an artificially dry baseline, and a single baseline per participant was applied to all sites.
5. The 1 h readings were confounded by surface film, and the first reading time (0.5 h in the protocol, 1 h in the data sheets) needs confirming.
6. The chyura butter used was only partly refined (acid value 5.6); fully refined chyura might perform differently.
7. Only two concentrations were tested, so the level at which chyura itself stops improving is not known.



## 11. Conclusion

In this project I compared chyura, shea and kokum butter using standard physico-chemical tests and then tested all three as the emollient in an oil-in-water moisturiser on skin. All three have melting points at or just above skin temperature (32.5–38 °C), normal saponification values and low peroxide values, so all three are suitable fats for beauty products. Chyura has the lowest iodine value (39.75) and the lowest peroxide value of the three, which makes it the most stable, the highest melting point (38 °C), and a high palmitic acid content that gives it a soft, ghee-like texture. Its one weak point, an acid value of 5.6 mg KOH/g, is because the sample was only partly refined; refining brings it in line with the others. On skin, all six creams took the skin from "normal" to "well hydrated" within two hours and kept it there for six hours. At 6% the butters ranked chyura > shea > kokum, both at six hours (58.5, 56.6 and 56.0 units) and more clearly in cumulative hydration over the whole period, while at 4% the three were indistinguishable. Chyura was the only butter that hydrated better at 6% than at 4%, so 6% is its optimum level.

Shea keeps a real advantage in its anti-inflammatory and sun-protection compounds, and kokum is the hardest and best for summer products. But for Lotus Herbals, chyura offers something the other two cannot together: it is grown close to home, it is chemically stable, it hydrated skin at least as well as shea in the company's own test, it supports vulnerable Himalayan communities, it is good for the hills, and it gives the brand a new ingredient story. For these reasons, Lotus Herbals should make refined chyura butter, at about 6%, its first choice of emollient, use kokum alongside it where hardness is needed, and keep imported shea only for the few products where its special compounds matter. This could not only help the products benefit overall but also provide economic advantages like lower overall costs and profit along with an advantage to the broad community



## 12. Bibliography


1. Devkota, H.P. et al. (2019). Chyuri (Diploknema butyracea) Butter. In: Fruit Oils: Chemistry and Functionality. Springer.
2. Timsina et al. (2026). The "Butter Tree" (Diploknema butyracea) for Sustainable Food Systems and Functional Applications. Plant-Environment Interactions.
3. Sarruf, F.D. et al. (2020). Butyrospermum parkii butter increased the photostability and in vivo SPF of a molded sunscreen system. Journal of Cosmetic Dermatology.
4. Akihisa, T. et al. (2010). Anti-inflammatory and chemopreventive effects of triterpene cinnamates and acetates from shea fat. Journal of Oleo Science.
5. Panda, L.R.L. et al. (2024). Diploknema butyracea (Chyura): A viable livelihood option for hill communities of Uttarakhand. Biological Forum, 16(2).
6. ICIMOD (2019). Community based chyura and beekeeping enterprise improves rural livelihoods. www.icimod.org
7. Kew Gardens / IUCN Red List. Vitellaria paradoxa (Shea tree): Vulnerable.

Tridge. Global shea butter price data, 2022